SUMMARY AND RECOMENDATION

1. Service-Based Churn Analysis

The type of service a customer subscribes to is the strongest predictor of their likelihood to leave.

Service Category	High Churn Segment	Churn Rate (%)	Key Insight
Internet Service	**Fiber Optic**	**~42%**	Fiber optic users churn at more than double the rate of DSL users (~19%).
Online Security	**No Service**	**~41%**	Customers without security are nearly 3x more likely to churn than those with it (~14%).
Tech Support	**No Service**	**~41%**	Lack of technical support is a major friction point for retention.
Contract Type	**Month-to-Month**	**~43%**	Short-term contracts account for the vast majority of total churn.

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2. Critical Demographic & Financial Drivers

Beyond the services themselves, how and who is paying significantly impacts the data.

- **Payment Method (Electronic Check):** This segment shows a churn rate of approximately **45%**. In contrast, customers using automated methods (Credit Card or Bank Transfer) show churn rates below **16%**.
- **Senior Citizens:** While a smaller portion of the total base, Senior Citizens have a higher churn probability (**~25%**) compared to younger demographics.
- **Tenure Impact:** There is a heavy concentration of churn (over **50% of total churners**) within the first **6 months** of joining. Once a customer passes the 24-month mark, the churn rate drops below **10%**.

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3. The "Sticky" vs. "Volatile" Services

Your analysis highlights a clear divide between services that anchor a customer and those that do not.

High-Retention "Sticky" Services

These services show a **negative correlation** with churn (meaning having them keeps customers):

- **Online Security & Backup:** Combined, these reduce the churn probability by over **60%** compared to baseline users.

- **Two-Year Contracts:** These customers have a retention rate of over **97%**.

Low-Impact "Volatile" Services

These services show a **neutral or positive correlation** with churn:

- **Streaming TV/Movies:** These have a churn rate of **~30%**, which is actually slightly higher than the average. This suggests entertainment is not a primary driver of loyalty.
- **Multiple Lines:** The churn rate for those with multiple lines (**~28%**) is nearly identical to those with a single line (**~25%**).

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4. Strategic Summary for Analysis

To optimize retention, the data suggests focusing on the **"High-Risk Trio"**:

1. **Fiber Optic Users** (42% Churn)
2. **Month-to-Month Contracts** (43% Churn)
3. **Electronic Check Users** (45% Churn)

Recommendation: Encouraging Fiber Optic users to add **Tech Support** or **Online Security** could potentially reduce their churn rate from **42% down to roughly 15%**, based on the performance of those sub-segments in your data.